

All-Order Tensor Inverse Jets for Multidimensional Pandrosion

Multilinear normal forms, m -flat charts, and anchor-order multiplier suppression

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Abstract

The scalar higher-order Pandrosion theory shows that local inverse jets flatten an analytic equation and make the fixed-anchor multiplier scale like a high power of the anchor distance. The tensorial multidimensional paper proved the first nontrivial case: a quadratic inverse jet cancels the bilinear residual tensor of an analytic system. This paper gives the all-order version. Let $G : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be analytic near a regular zero ρ , with $A = DG(\rho)$ invertible. For every $m \geq 1$, there is a unique polynomial jet

$$\phi_m(u) = \rho + B_1(u) + B_2(u, \dots, u) + \dots + B_m(u, \dots, u)$$

whose homogeneous symmetric multilinear coefficients satisfy

$$G(\phi_m(u)) = u + \mathcal{O}(\|u\|^{m+1}).$$

The coefficients are determined recursively: at each order only AB_k is unknown, and all lower terms are already fixed. Thus ϕ_m is the degree- m Taylor jet of the local inverse of G . For the charted averaged-Jacobian Pandrosion map, this gives

$$\|DP_{G \circ \phi_m, a}(0)\| = \mathcal{O}(\|a\|^m).$$

More sharply, if the first uncanceled defect is a homogeneous tensor H_{m+1} , then

$$DP_{G \circ \phi_m, a}(0) = \frac{1}{m+1} DH_{m+1}(a) + \mathcal{O}(\|a\|^{m+1}).$$

The paper closes with a finite-probe reconstruction conjecture: a stable Pandrosion engine that estimates multilinear defects from symmetric probes should recover the inverse-jet hierarchy asymptotically.

Scope and status. This is a theorem-and-conjecture manuscript inside the Pandrosion research programme. The theorems are local analytic normal-form statements; the final reconstruction statement is conjectural and meant to be tested against finite-slope engines.

1 From tensor-flat charts to all orders

Earlier scalar papers established that the anchored divided-difference map improves when the equation is first expressed in a chart where the transformed residual is nearly linear. The multidimensional tensor paper identified the leading obstruction:

$$F(u) = u + \frac{1}{2}T_2[u, u] + \dots \implies P_{F,a}(s) = \frac{1}{2}T_2[a, s] + \dots$$

Cancelling T_2 by a quadratic inverse jet makes the multiplier second order in the anchor distance. The natural continuation is to cancel every homogeneous tensor up to a prescribed order.

2 Homogeneous tensor jets

Let G be analytic near $\rho \in \mathbb{C}^n$, $G(\rho) = 0$, and $A = DG(\rho)$ invertible. Write the Taylor expansion

$$G(\rho + w) = Aw + \sum_{q \geq 2} G_q[w, \dots, w], \quad (1)$$

where $G_q \in \text{Sym}^q((\mathbb{C}^n)^*) \otimes \mathbb{C}^n$ is the symmetric q -linear coefficient $G_q = D^q G(\rho)/q!$.

Definition 2.1 (Polynomial inverse jet). *An order- m tensor jet at ρ is a polynomial map*

$$\phi_m(u) = \rho + \sum_{k=1}^m B_k[u, \dots, u],$$

where $B_k \in \text{Sym}^k((\mathbb{C}^n)^*) \otimes \mathbb{C}^n$. It is m -flat for G if

$$G(\phi_m(u)) = u + \mathcal{O}(\|u\|^{m+1}).$$

Theorem 2.2 (All-order tensor inverse jet). *For every $m \geq 1$, there is a unique order- m tensor jet ϕ_m such that*

$$G(\phi_m(u)) = u + \mathcal{O}(\|u\|^{m+1}).$$

The coefficients are determined recursively. First $B_1 = A^{-1}$. If $B_1, \dots, B_{k-1}$ are known, let $R_k(u)$ be the homogeneous degree- k part of

$$\sum_{q=2}^k G_q[W_{k-1}(u), \dots, W_{k-1}(u)], \quad W_{k-1}(u) = \sum_{j=1}^{k-1} B_j[u, \dots, u].$$

Then

$$B_k[u, \dots, u] = -A^{-1}R_k(u).$$

Moreover, ϕ_m is exactly the degree- m Taylor jet of the local inverse of G at 0.

Proof. The inverse function theorem gives a local inverse Ψ with $G(\Psi(u)) = u$, so existence of a formal Taylor jet is guaranteed. The recursive proof also gives uniqueness. For $k = 1$, the degree-one term in $G(\phi_m(u))$ is AB_1u , hence $B_1 = A^{-1}$.

Assume $B_1, \dots, B_{k-1}$ have been chosen so that all homogeneous terms of degrees $2, \dots, k-1$ vanish. In degree k , the only term involving the unknown B_k is $AB_k[u, \dots, u]$, because every nonlinear G_q has $q \geq 2$ and therefore uses only the already known lower coefficients to contribute to degree k . The remaining degree- k contribution is $R_k(u)$. The equation

$$AB_k[u, \dots, u] + R_k(u) = 0$$

has the unique solution stated above because A is invertible. Induction constructs a unique jet through order m , and it must agree with the Taylor truncation of the analytic inverse. $\square$

Remark 2.3. For $m = 2$, [Theorem 2.2](#) gives

$$\phi_2(u) = \rho + A^{-1}u - A^{-1}G_2[A^{-1}u, A^{-1}u],$$

which is the quadratic tensor-flat chart from the multidimensional Pandrosion paper.

3 Averaged-Jacobian Pandrosion in an m -flat chart

Let $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be analytic near 0, $F(0) = 0$, and $DF(0) = I$. The averaged-Jacobian divided difference is

$$\mathcal{Q}_F(a, s) = \int_0^1 DF(s + t(a - s)) dt,$$

so that $F(a) - F(s) = \mathcal{Q}_F(a, s)(a - s)$. When $\mathcal{Q}_F(a, s)$ is invertible, the anchored multidimensional Pandrosion map is

$$P_{F,a}(s) = a - \mathcal{Q}_F(a, s)^{-1}F(a) = s - \mathcal{Q}_F(a, s)^{-1}F(s).$$

Definition 3.1 (m -flat residual). *The residual F is m -flat at 0 if*

$$F(u) = u + \mathcal{O}(\|u\|^{m+1}).$$

If

$$F(u) = u + H_{m+1}(u) + \mathcal{O}(\|u\|^{m+2}),$$

with H_{m+1} homogeneous of degree $m+1$, then H_{m+1} is the leading defect tensor.

Theorem 3.2 (Anchor-order multiplier law). *Assume F is m -flat at 0. Then, for small anchors a with $\mathcal{Q}_F(a, 0)$ invertible,*

$$\|DP_{F,a}(0)\| = \mathcal{O}(\|a\|^m).$$

If the leading defect is H_{m+1} , then

$$DP_{F,a}(0) = \frac{1}{m+1}DH_{m+1}(a) + \mathcal{O}(\|a\|^{m+1}). \quad (2)$$

Proof. Using the secant form of the map and $F(0) = 0$,

$$DP_{F,a}(0) = I - \mathcal{Q}_F(a, 0)^{-1}DF(0) = I - \mathcal{Q}_F(a, 0)^{-1}.$$

Since $F(u) = u + \mathcal{O}(\|u\|^{m+1})$, we have $DF(u) = I + \mathcal{O}(\|u\|^m)$, hence

$$\mathcal{Q}_F(a, 0) = \int_0^1 DF(ta) dt = I + \mathcal{O}(\|a\|^m).$$

This proves the norm estimate. If $F(u) = u + H_{m+1}(u) + \mathcal{O}(\|u\|^{m+2})$, then

$$DF(ta) = I + t^m DH_{m+1}(a) + \mathcal{O}(\|a\|^{m+1}),$$

so

$$\mathcal{Q}_F(a, 0) = I + \frac{1}{m+1}DH_{m+1}(a) + \mathcal{O}(\|a\|^{m+1}).$$

Expanding the inverse gives (2). □

Corollary 3.3 (All-order charted Pandrosion). *Let $G(\rho) = 0$ and $DG(\rho)$ be invertible. Let ϕ_m be the all-order tensor inverse jet from [Theorem 2.2](#), and set $F_m = G \circ \phi_m$. Then*

$$\|DP_{F_m,a}(0)\| = \mathcal{O}(\|a\|^m).$$

Proof. By construction, $F_m(u) = u + \mathcal{O}(\|u\|^{m+1})$. Apply [Theorem 3.2](#). □

4 Two-point residual estimate

Proposition 4.1 (Uniform two-point order). *If $F(u) = u + \mathcal{O}(\|u\|^{m+1})$, then for a, s sufficiently small,*

$$\|P_{F,a}(s)\| \leq C \|s\| (\|a\| + \|s\|)^m.$$

Proof. Write $F(s) = s + R(s)$, with $\|R(s)\| \leq C_0 \|s\|^{m+1}$. Also

$$\mathcal{Q}_F(a, s) = I + E(a, s), \quad \|E(a, s)\| \leq C_1(\|a\| + \|s\|)^m.$$

For a, s small, $\mathcal{Q}_F(a, s)^{-1} = I - E(a, s) + \mathcal{O}(\|E\|^2)$. Therefore

$$P_{F,a}(s) = s - \mathcal{Q}_F(a, s)^{-1}F(s) = E(a, s)s - R(s) + \mathcal{O}(\|s\| \|E(a, s)\|^2),$$

which gives the claimed bound after shrinking the neighborhood. $\square$

5 Finite-probe reconstruction conjecture

The theorems above use derivatives to define the inverse jet. The practical Pandrosion programme uses finite probes. The bridge is a reconstruction conjecture.

Conjecture 5.1 (Finite-probe inverse-jet reconstruction). *Let G be analytic near a regular zero. Suppose a Pandrosion engine uses symmetric probe clouds whose radius tends to zero, rejects rank-defective finite-slope matrices, and estimates homogeneous defects by multilinear polarization identities. Then the estimated chart coefficients converge order by order to the tensor inverse-jet coefficients $B_1, \dots, B_m$ of [Theorem 2.2](#).*

Remark 5.2. *This conjecture is deliberately algorithmic. The theorem says what the optimal tensors are; the conjecture says that a finite-slope engine can recover them without direct derivative access.*

6 Conclusion

The all-order tensor inverse jet is the multidimensional normal form behind Pandrosion chart optimization. It unifies scalar higher-order cancellation, quadratic tensor-flat charts, and the atlas programme: every regular zero has a canonical hierarchy of local coordinates in which the raw anchored multiplier is suppressed from $\mathcal{O}(\|a\|)$ to $\mathcal{O}(\|a\|^m)$. The next step is to make these local charts uniform across a finite atlas.

References

- [1] I. Besevic, *Anchored Pandrosion for General Analytic Equations*, manuscript, May 2026.
- [2] I. Besevic, *Higher-Order Jet Cancellation for Anchored Divided-Difference Iterations*, manuscript, May 2026.
- [3] I. Besevic, *Tensorial Multidimensional Pandrosion*, manuscript, May 2026.
- [4] I. Besevic, *Stable Numerical Atlases for Pandrosion Dynamics*, manuscript, May 2026.